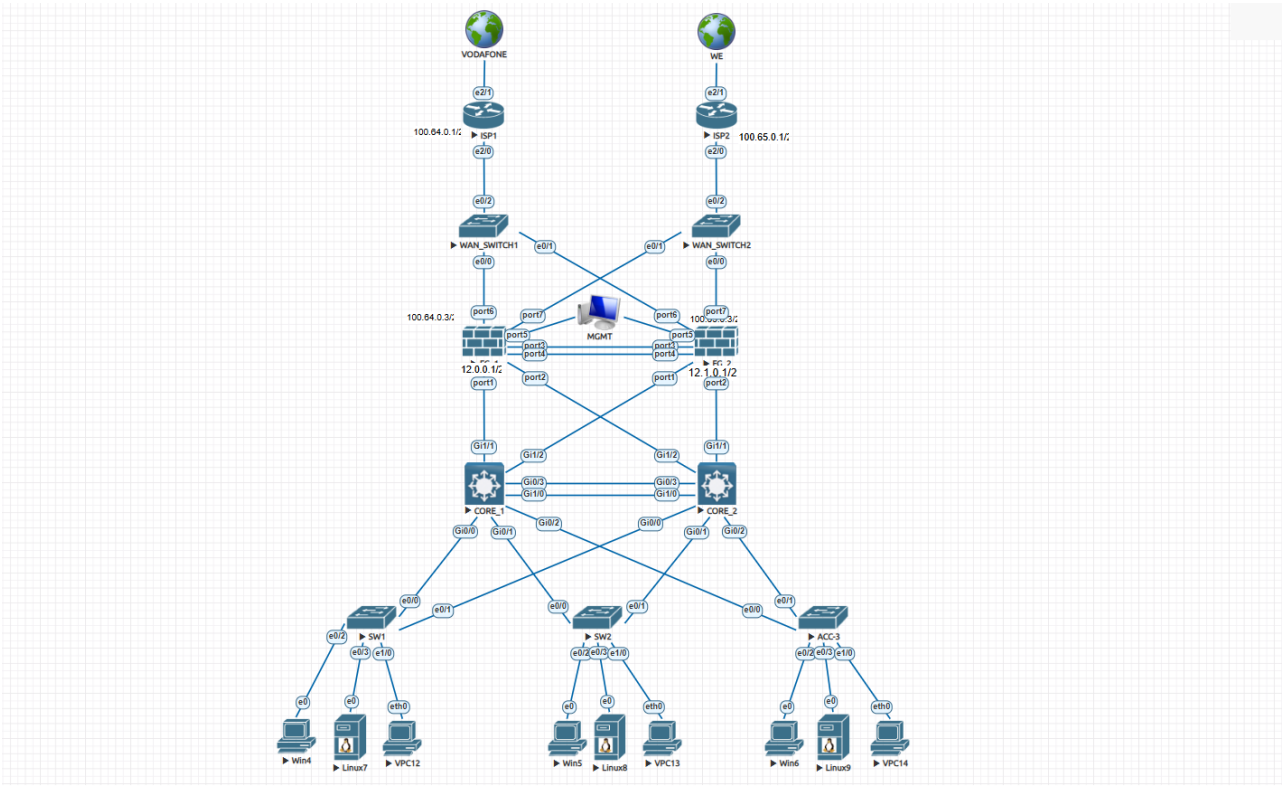
Enterprise Network with HA FortiGate Firewalls and Dual-ISP SD-WAN

Project Explained, Step by Step



Network topology (EVE-NG)

EVE-NG Lab | FortiGate HA | SD-WAN | OSPF | HSRP | MSTP

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Chapter 1: The Project in Simple Terms

This project is a **company network** designed so that **if any device or link fails, the network keeps working** and employees continue to reach the internet without noticing.

> **Simple analogy**
> Picture an office building with **two main entrances** to the internet (Vodafone and WE), **two security guards** at the door (the FortiGate firewalls), and **two distribution managers** on the floor below (the core switches). Every department reaches the managers through two different paths.
>
> If an entrance, a guard or a manager is out, the other one takes over immediately.

Goals

- No single point of failure: ISP, firewall, core and access layers are all duplicated.
- The core and the firewall exchange routes automatically using **OSPF**.
- Both cores work at the same time (not one active and one idle) thanks to **HSRP** and **MSTP**.
- If an ISP fails, traffic moves to the other one automatically thanks to **SD-WAN**.
- Internet access is controlled with **NAT**, a firewall policy and web filtering.

What protects against what?

Layer	Devices	Solution	If this happens
Internet	ISP1 (Vodafone), ISP2 (WE)	SD-WAN with health checks	ISP outage or link cut
WAN switches	WAN_SWITCH1, WAN_SWITCH2	Each firewall connects to both	One WAN switch fails
Firewall	FG-1, FG-2	Active-Passive HA	A firewall fails
Core	NH_PRIM, NH_SEC	HSRP, OSPF, transit VLAN per core	A core fails
Access	ACC-1, ACC-2, ACC-3	Two uplinks per switch, MSTP	An uplink or a core fails

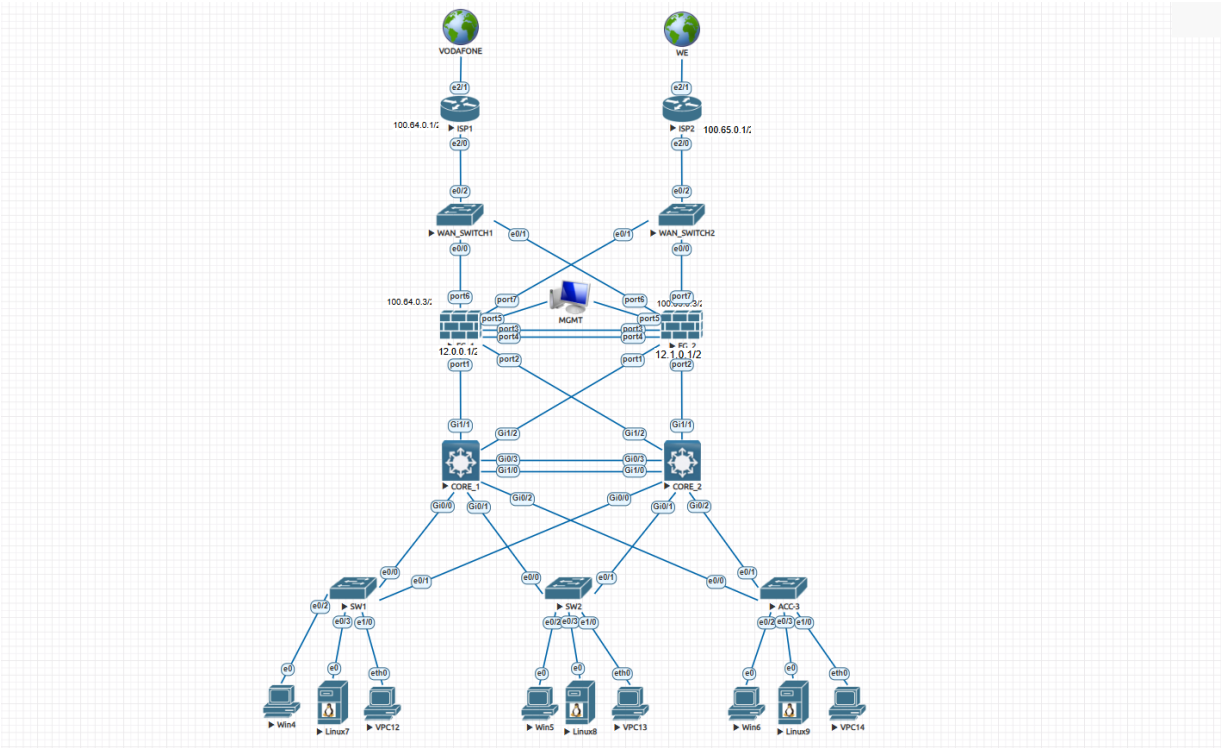
Devices and platform

Type	Details
Platform	EVE-NG
Firewalls	2 x FortiGate VM64-KVM (FortiOS 6.4.6)
Core	2 x Cisco L3 switches: NH_PRIM (CORE_1) and NH_SEC (CORE_2)

Type	Details
Access	3 x Cisco L2 switches: ACC-1 (SW1), ACC-2 (SW2), ACC-3
WAN	2 x Cisco routers (ISP1, ISP2) and 2 x L2 WAN switches
End devices	Windows, Linux and VPCS hosts in VLANs 10 to 50
Protocols	OSPF, HSRP, MSTP, EtherChannel, SD-WAN, NAT

Chapter 2: Network Topology

This is the whole network from EVE-NG. Read it from **top to bottom**:



Full network topology in EVE-NG

There are four layers:

- **Layer 1, Internet:** VODAFONE and WE (the globe icons are the EVE-NG cloud). Each connects to a router (ISP1 or ISP2) and then to a switch (WAN_SWITCH1 or WAN_SWITCH2).
- **Layer 2, Firewall:** FG_1 and FG_2 run as an HA pair. Ports 6 and 7 face the internet, ports 1 and 2 face the internal network.
- **Layer 3, Core:** CORE_1 and CORE_2, connected to each other by two links.
- **Layer 4, Access:** SW1, SW2 and ACC-3. Each connects to both cores and carries the end devices (Win, Linux, VPC).

FortiGate ports: what each one does

Port	Purpose	Connected to
port1	Internal network (LAN)	CORE_1 (from both firewalls)
port2	Internal backup (LAN_BACKUP)	CORE_2 (from both firewalls)
port3, port4	HA heartbeat	Two direct cables FG_1 to FG_2

Port	Purpose	Connected to
port5	Management (MGMT)	Management host
port6	Primary internet (WAN)	WAN_SWITCH1 to ISP1 (Vodafone)
port7	Backup internet (WAN_BACKUP)	WAN_SWITCH2 to ISP2 (WE)

Chapter 3: IP Addressing and VLAN Plan

Every network needs an addressing plan. This is the complete plan for this project:

User VLANs

VLAN	Name	Subnet	Gateway (VIP)	Preferred core
10	PCs / Users	10.10.10.0/24	10.10.10.254	NH_PRIM
20	IP Phones	10.20.20.0/24	10.20.20.254	NH_PRIM
30	Wi-Fi / WLAN	10.30.30.0/24	10.30.30.254	NH_SEC
40	Management	10.40.40.0/24	10.40.40.254	NH_SEC
50	Servers	10.50.50.0/24	10.50.50.254	NH_SEC

What is the VIP .254?

It is a **virtual** address that every device in the VLAN uses as its default gateway. The active core answers it. If that core fails, the other core answers the same address, so end devices never change their settings.

Links between devices

Link	Subnet	Addresses
FortiGate port1 to NH_PRIM (VLAN 12)	12.0.0.0/29	FortiGate .1 (cluster IP), core .2
FortiGate port2 to NH_SEC (VLAN 13)	12.1.0.0/29	FortiGate .1 (cluster IP), core .2
WAN 1: Vodafone	100.64.0.0/29	ISP1 .1, FortiGate port6 .3
WAN 2: WE	100.65.0.0/29	ISP2 .1, FortiGate port7 .3
HA heartbeat	port3 and port4	Direct cables between the firewalls
Firewall management (port5)	192.168.1.0/24	FG-1 .52, FG-2 .53

What does /29 mean?

A /29 subnet has **8 addresses, of which 6 are usable** (.1 to .6). It is a good fit for links between devices, because those links only have two or three devices. A /24 gives **254 hosts** and is used for the VLANs.

Important

Both FortiGates share the **same addresses**. Whichever unit is Active owns them. If it fails, the other unit takes the same addresses and the same MAC.

Chapter 4: The Access Layer

Simple analogy

A VLAN is like **company departments**: one for employees, one for phones, one for servers. Each department has its own network even if they all plug into the same switch.

Devices on each switch

Each access switch has three devices, each in a different VLAN:

Switch	Et0/2 (Windows)	Et0/3 (Linux)	Et1/0 (VPCS)
ACC-1 (SW1)	VLAN 10 - Win4	VLAN 20 - Linux7	VLAN 30 - VPC12
ACC-2 (SW2)	VLAN 10 - Win5	VLAN 20 - Linux8	VLAN 40 - VPC13
ACC-3	VLAN 10 - Win6	VLAN 20 - Linux9	VLAN 50 - VPC14

Uplinks

- Et0/0 goes to **CORE_1** and Et0/1 goes to **CORE_2**.
- Both are **trunks**, so all VLANs travel over one cable.
- If one uplink fails, the other keeps working.

Protection on host ports

- **Port security (sticky):** the switch remembers the MAC address of the first device plugged in. If a different device is connected, the port shuts down.
- **PortFast:** the port comes up immediately instead of waiting several seconds, because it connects to a host and not a switch.

Lesson learned

PortFast of type **network** enables Bridge Assurance, and the switch waits for BPDUs from the other side. Normal hosts never send BPDUs, so the switch blocks the port (state BKN). On host ports always use the regular PortFast (**edge**).

Chapter 5: The Core Layer: HSRP, MSTP and EtherChannel

The core is the heart of the network. Three features work together.

First: HSRP, one gateway on two switches

> **Simple analogy**
> HSRP is like a **shared phone number** for a department. People call the number (.254) and it does not matter which core answers. If the one answering fails, the other one answers.

VLAN	Active core	Standby core
10 and 20	NH_PRIM (priority 110 with preempt)	NH_SEC
30, 40 and 50	NH_SEC (priority 110 with preempt)	NH_PRIM

Second: MSTP, loop prevention and load sharing

When every switch connects to both cores, a **loop** can form (a packet circulates forever). Spanning Tree blocks one link and opens it again if the other one fails. With **MSTP** we split the VLANs into two groups (instances):

- **Instance 1** (VLANs 10 and 20): root bridge is **NH_PRIM** (priority 24576), backup NH_SEC (28672).
- **Instance 2** (VLANs 30, 40 and 50): root bridge is **NH_SEC** (priority 24576), backup NH_PRIM (28672).

Result: **both cores are active**, each serving certain VLANs, and neither is idle. The region name is **SPMST**.

Third: EtherChannel between the cores

Two links (Gi0/3 and Gi1/0) between NH_PRIM and NH_SEC are bundled into **Port-channel1**, behaving like one faster link. It is a trunk for VLANs 10 to 50. If one link fails, the other continues.

Chapter 6: Core to Firewall Connection and OSPF

Why transit VLANs 12 and 13?

Each core connects to **both** firewalls (FG_1 and FG_2). To put both firewalls on the **same network** from the core's point of view, we placed the two ports in a **dedicated VLAN** and created an **SVI** (an IP address on the core):

Core	VLAN	SVI address	Ports	Connected to
NH_PRIM (CORE_1)	12	12.0.0.2/29	Gi1/1, Gi1/2 (access)	port1 on FG-1 and FG-2
NH_SEC (CORE_2)	13	12.1.0.2/29	Gi1/1, Gi1/2 (access)	port2 on FG-1 and FG-2

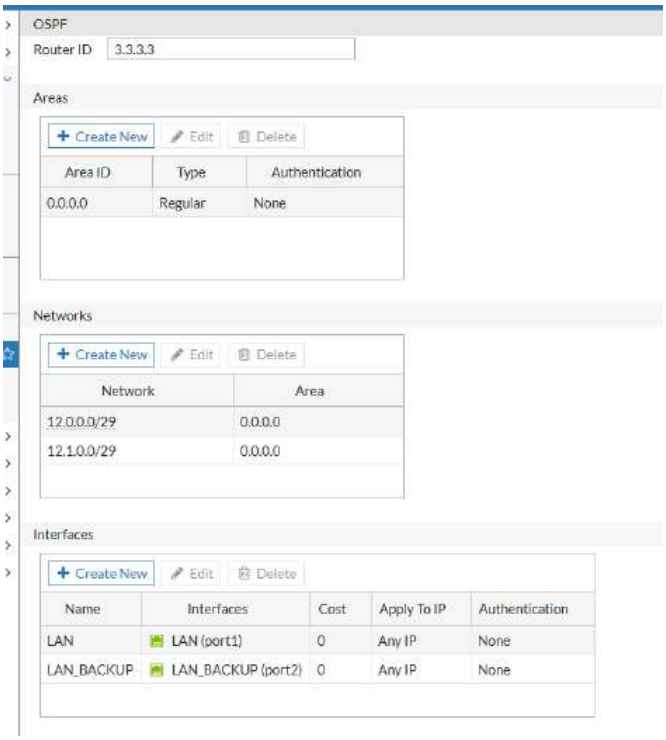
> **The secret**
> The **Active** FortiGate holds the address 12.0.0.1. If it fails, the other one takes **the same address and the same MAC**, so the core notices nothing and no routing change is needed.

OSPF in simple terms

OSPF is a protocol that lets each device tell its neighbors: "I know these networks." Routes are then built **automatically** instead of being typed by hand.

- The cores advertise VLANs 10 to 50 to the firewall.
- The firewall advertises that it is the **gateway to the internet** (the default route 0.0.0.0/0) using default-information-originate always.
- The cores run OSPF **only on the transit SVI** (all other interfaces are passive).
- On the FortiGate, OSPF runs on **port1 and port2 only**. The WAN ports (6 and 7) are **not** in OSPF.

Device	OSPF Router-ID	Area
NH_PRIM	1.1.1.1	0
NH_SEC	2.2.2.2	0
FortiGate	3.3.3.3	0



OSPF settings on the FortiGate: networks and the LAN and LAN_BACKUP interfaces

Verification

On the FortiGate:

```
FG-1 # get router info ospf neighbor
Neighbor ID  Pri  State        Dead Time  Address    Interface
1.1.1.1        1  Full/Backup  00:00:32   12.0.0.2   port1
2.2.2.2        1  Full/Backup  00:00:37   12.1.0.2   port2
```

On NH_PRIM:

```
NH_PRIM# show ip route ospf
Gateway of last resort is 12.0.0.1 to network 0.0.0.0
O*E2  0.0.0.0/0 [110/10] via 12.0.0.1, Vlan12
```

How to read the output

Full means a complete, healthy OSPF relationship with the neighbor. Seeing both cores (1.1.1.1 and 2.2.2.2) means the firewall sees both of them.

The line **O*E2 0.0.0.0/0** on the core means the core learned the default route from the firewall.

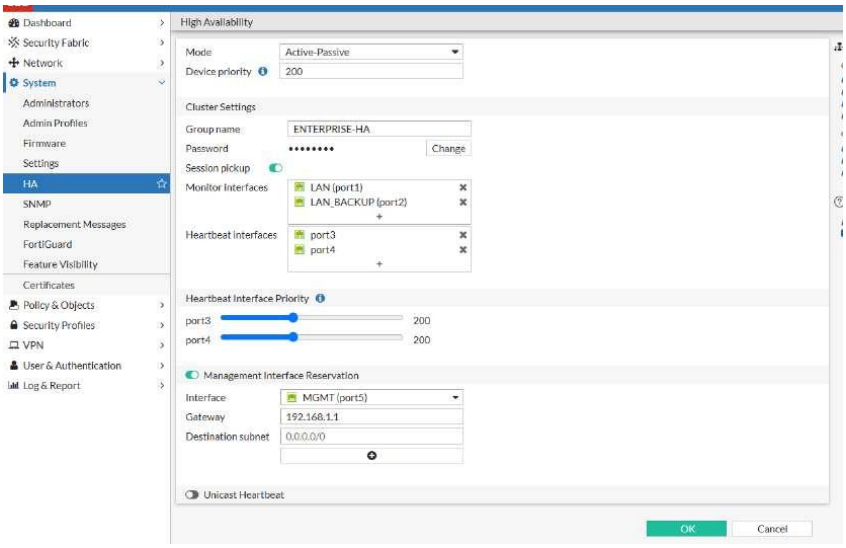
Chapter 7: FortiGate High Availability

Simple analogy

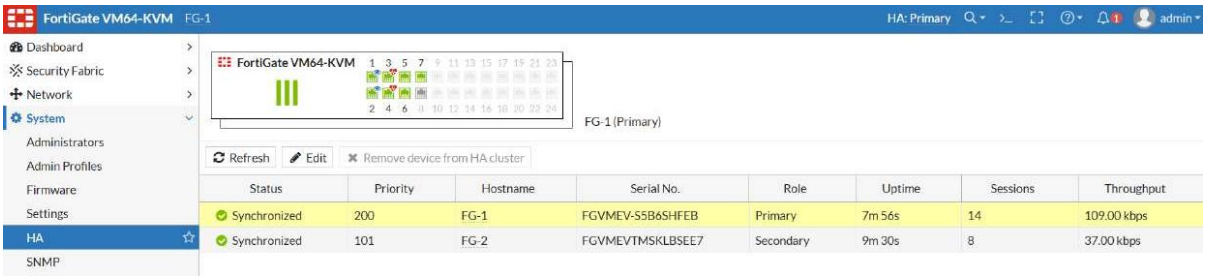
Two guards: one is **on duty** (Primary) and the other **stands beside him** watching (Secondary). If the first one falls, the second takes his place **with the same addresses** and nobody notices.

Settings

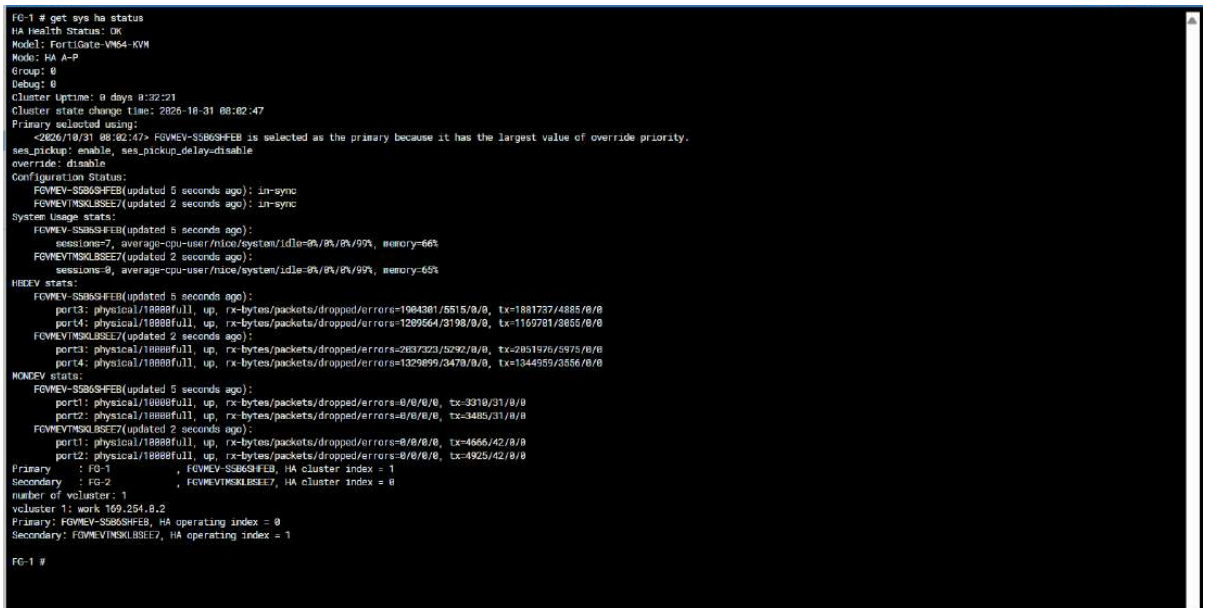
Setting	Value	Meaning
Mode	Active-Passive	One unit works, the other is standby
Group name	ENTERPRISE-HA	Name of the cluster
Priority	FG-1 = 200, FG-2 = 101	The higher value becomes Primary
Heartbeat	port3 and port4	Two cables the firewalls use to exchange keep-alives
Session pickup	Enabled	Established connections survive the failover
Monitored interfaces	port1 and port2	If one fails, a failover happens
Management interface	port5 (gateway 192.168.1.1)	Keeps each unit reachable



HA settings on FG-1



HA monitor: FG-1 is Primary, FG-2 is Secondary, both Synchronized



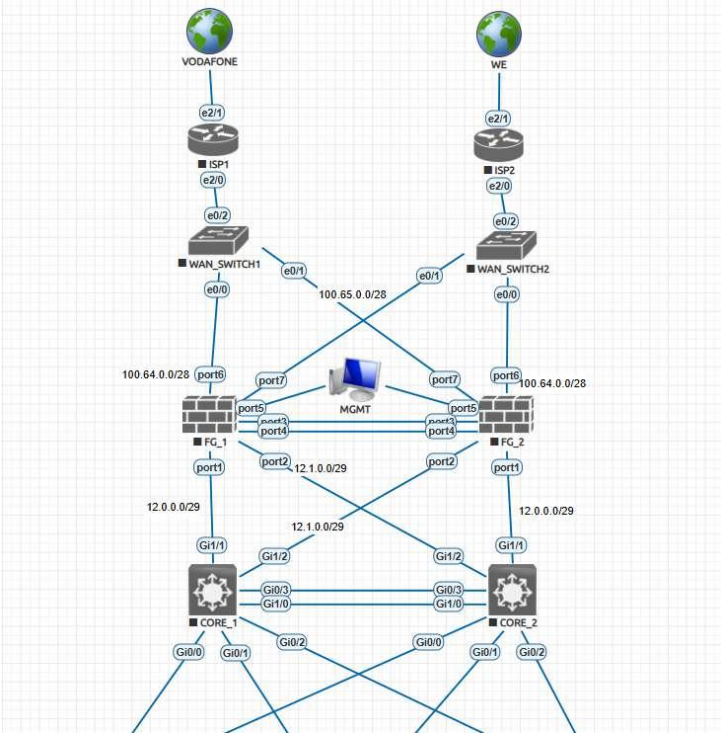
Output of get system ha status: health OK and both units in-sync

The cabling rule

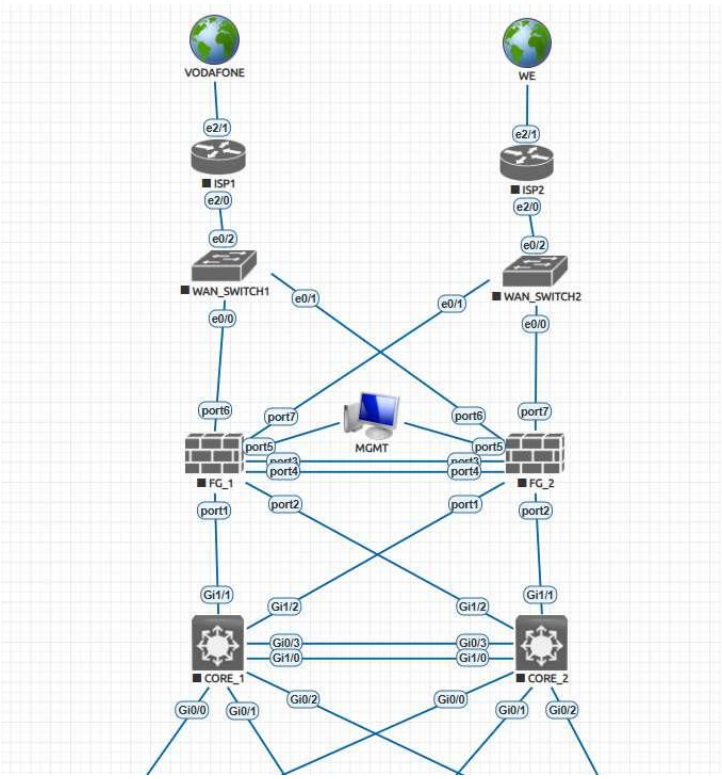
HA **synchronizes** configuration between the firewalls, so **port6 on both units has the same IP**. This is why **the same port on both firewalls must connect to the same switch**:

- port6 on FG_1 and FG_2 goes to WAN_SWITCH1 (Vodafone).
- port7 on FG_1 and FG_2 goes to WAN_SWITCH2 (WE).
- port1 on both goes to CORE_1, and port2 on both goes to CORE_2.

At first the cabling was **crossed** (FG_2 port6 went to WAN_SWITCH2). On the first failover, the Vodafone IP would have appeared on the WE switch and the internet would have gone down.



Before: crossed cabling (FG_2 port6 to WAN_SWITCH2)



After: the same port on both firewalls goes to the same switch

Chapter 8: The Internet Side: ISP1 and ISP2

Each ISP is simulated by a Cisco router that does three things:

- **Firewall side (e2/0):** a /29 address (ISP1 = 100.64.0.1, ISP2 = 100.65.0.1), configured as NAT Inside.
- **Internet side (e2/1):** gets an address by **DHCP** from the cloud (NAT Outside).
- **NAT overload:** translates internal addresses to the e2/1 address.

The **WAN switches** are Layer 2 only (VLAN 100), with no IP and no routing. Each has three ports: the first firewall, the second firewall and the ISP.

NAT happens twice

Step	Device	Source after translation
1	A host (for example in VLAN 50)	10.50.50.x
2	FortiGate (NAT)	100.64.0.3 (the port6 address)
3	ISP1 (NAT overload)	The e2/1 address (for example 192.168.1.56)

```
ISP1#sh ip nat translations
Pro Inside global      Inside local       Outside local      Outside global
icmp 192.168.1.56:1420 100.64.0.3:1420    8.8.8.8:1420       8.8.8.8:1420
icmp 192.168.1.56:1421 100.64.0.3:1421    1.1.1.1:1421       1.1.1.1:1421
udp 192.168.1.56:2907  100.64.0.3:2907    208.91.112.52:53   208.91.112.52:53
udp 192.168.1.56:2907  100.64.0.3:2907    208.91.112.53:53   208.91.112.53:53
tcp 192.168.1.56:11416 100.64.0.3:11416   173.243.143.6:443  173.243.143.6:443
tcp 192.168.1.56:11419 100.64.0.3:11419   173.243.143.6:443  173.243.143.6:443
tcp 192.168.1.56:11420 100.64.0.3:11420   173.243.143.6:443  173.243.143.6:443
tcp 192.168.1.56:11421 100.64.0.3:11421   173.243.143.6:443  173.243.143.6:443
tcp 192.168.1.56:13355 100.64.0.3:13355   209.40.117.130:443 209.40.117.130:443
tcp 192.168.1.56:13365 100.64.0.3:13365   209.40.117.130:443 209.40.117.130:443
icmp 192.168.1.56:60417 100.64.0.3:60417  8.8.8.8:60417      8.8.8.8:60417
ISP1#
```

ISP1 NAT table: 100.64.0.3 is translated to the e2/1 address

How to read the NAT table

Inside local is the address before translation (100.64.0.3). **Inside global** is the address after translation (192.168.1.56). **Outside** is the destination (for example 8.8.8.8).

Chapter 9: SD-WAN: Automatic Failover Between the ISPs

Simple analogy

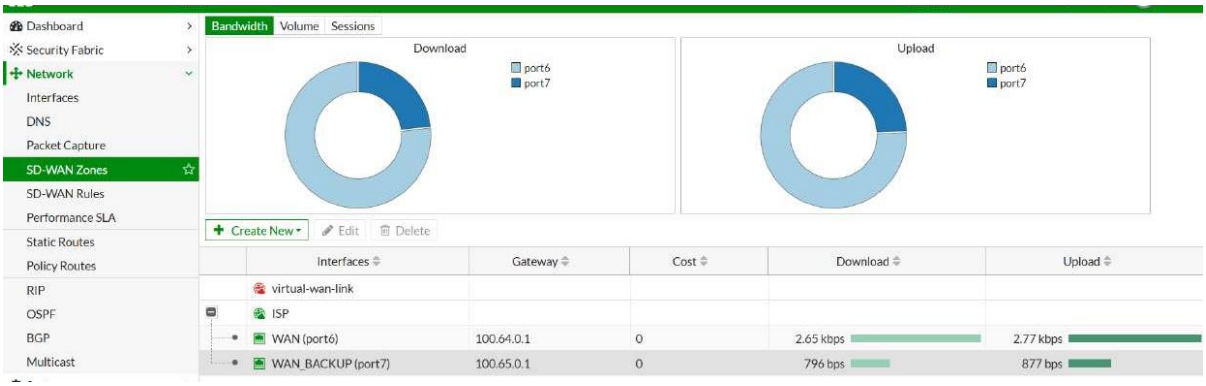
SD-WAN is like a **driver** with two roads to the office. Every few seconds he sends a test message down each road (the health check). If the first road is closed, he switches to the second one **by himself**.

Settings

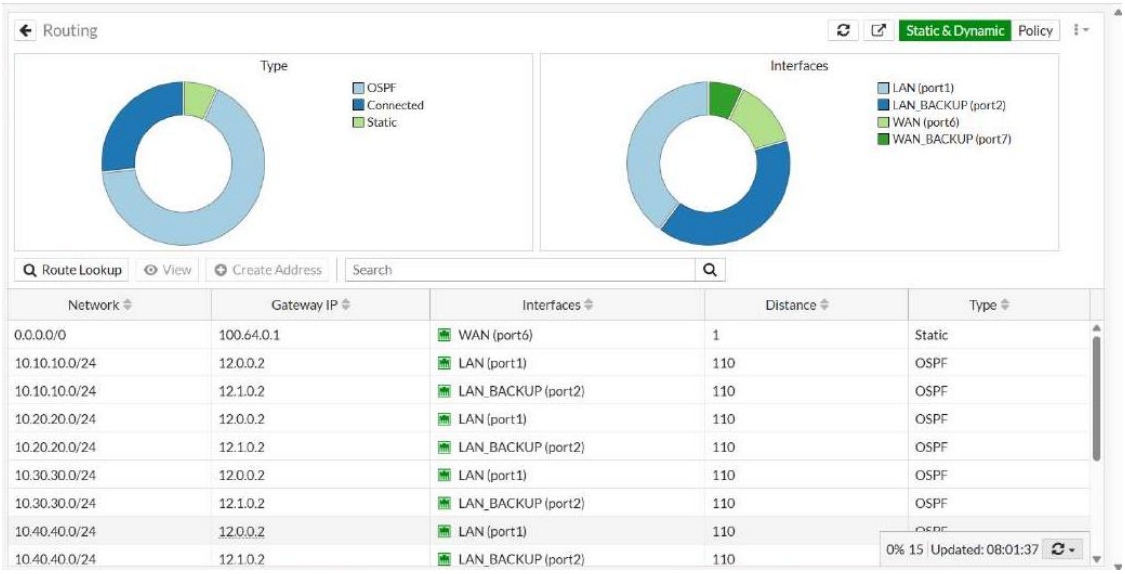
Part	Value
Zone	ISP
Member 1	port6 (Vodafone), gateway 100.64.0.1
Member 2	port7 (WE), gateway 100.65.0.1
Health check	Internet_Check: probes 8.8.8.8 and 1.1.1.1 every 1000 ms
Fail and recover	An ISP is declared dead after 5 failed probes and restored after 5 successful ones
SD-WAN rule	Primary_ISP1: member 1 is preferred, member 2 is the fallback
Static route	One default route (0.0.0.0/0) pointing to the SD-WAN

Order of implementation

1. Delete the old static routes (an interface that has a route cannot be added to SD-WAN).
2. Add port6 and port7 as members of a zone named ISP, with their gateways.
3. Create the Performance SLA (Internet_Check).
4. Create the SD-WAN rule: port6 first, then port7.
5. Create one default route on the SD-WAN.
6. Create the firewall policy (Chapter 10).



SD-WAN zone ISP with port6 and port7 and live bandwidth



Routing table: the default route and the routes learned from the cores

What does the routing table show?

0.0.0.0/0 through WAN (port6) of type Static is the default route toward the internet.

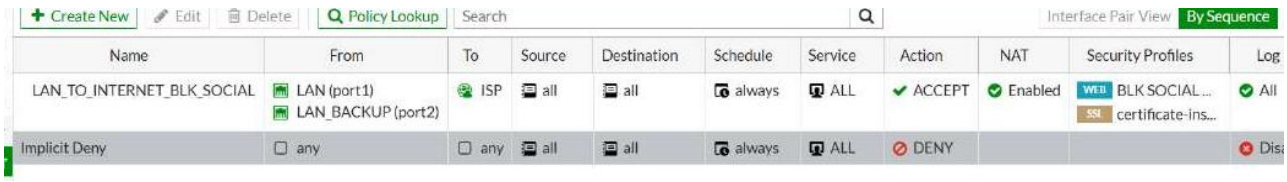
The **10.x.x.x** entries of type OSPF go through LAN (12.0.0.2) and LAN_BACKUP (12.1.0.2): the firewall sees every VLAN through both cores.

Chapter 10: Firewall Policy and Web Filter

The policy

The FortiGate blocks all traffic unless a policy allows it (the implicit deny). We have one policy for internet access:

Field	Value
Name	LAN_TO_INTERNET_BLK_SOCIAL
From	port1 (LAN) and port2 (LAN_BACKUP)
To	Zone ISP (SD-WAN)
Source / Destination / Service	all / all / ALL
Action	ACCEPT
NAT	Enabled (outgoing interface address)
SSL inspection	certificate-inspection
Logging	All sessions



Firewall policy list: the main policy and the implicit deny below it

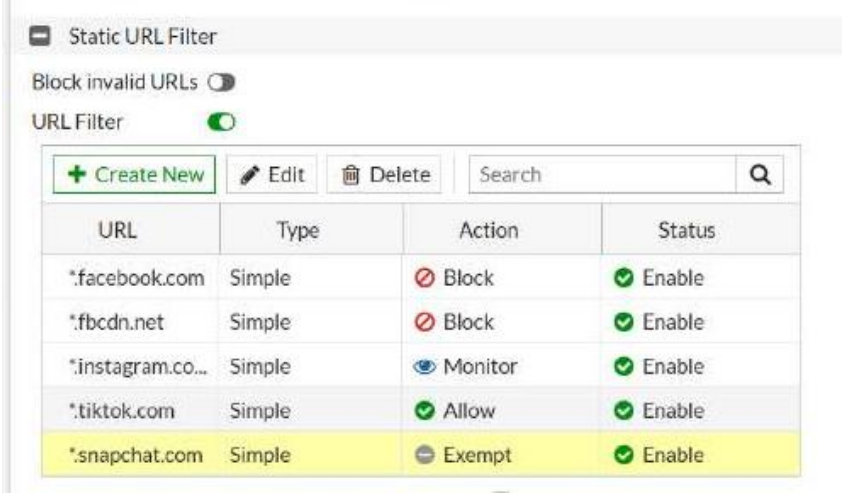
Why is NAT required?

The ISP cannot return replies to private 10.x.x.x addresses. With NAT, the firewall changes the source to its WAN address, so the reply comes back to it.

Web filter (blocking social media)

We used a **Static URL Filter**: a list of sites, each with an action:

Action	Meaning
Block	Blocks the site
Monitor	Allows and records in the log
Allow	Allows
Exempt	Allows and skips the other filters



Static URL filter: facebook.com and fbcdn.net are set to Block

Tip

Browsers (especially Chrome) may use **QUIC** over UDP/443, which the web filter does not see. To make blocking reliable, add a policy above the main one that blocks QUIC.

Chapter 11: Testing and Verification

We tested from two hosts in different VLANs:

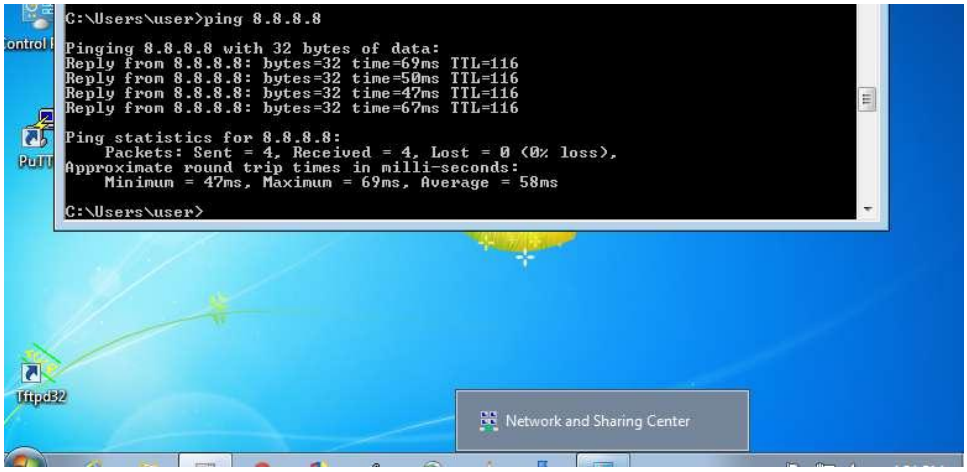
```
VPCS> ping 8.8.8.8

84 bytes from 8.8.8.8 icmp_seq=1 ttl=116 time=50.170 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=116 time=50.751 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=116 time=58.168 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=116 time=47.537 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=116 time=48.614 ms

VPCS> trace 8.8.8.8
trace to 8.8.8.8, 8 hops max, press Ctrl+C to stop
 1   10.50.50.2   14.830 ms  2.374 ms  1.528 ms
 2   12.1.0.1   15.603 ms  6.800 ms  5.264 ms
 3   100.64.0.1   14.801 ms  9.440 ms  10.222 ms
 4   192.168.1.1   19.948 ms  21.316 ms  18.198 ms
 5   100.122.0.1   19.202 ms  21.025 ms  20.314 ms
 6   10.29.93.177   19.619 ms  30.404 ms  30.432 ms
 7     *  *  *
 8     *  *  *

VPCS>
```

VPCS: ping to 8.8.8.8 succeeds and the trace shows the path



Windows: 4 replies from 8.8.8.8 with 0% loss

The path the packet takes (trace)

Hop	Address	Device
1	10.50.50.2	CORE_2 (gateway of VLAN 50)
2	12.1.0.1	FortiGate (cluster IP on port2)
3	100.64.0.1	ISP1 (Vodafone): SD-WAN chose port6
4	192.168.1.1	The real lab router
5 and beyond	100.122.0.1 ...	The real ISP network

What does TTL=116 mean?

TTL drops by one at every router. The reply from 8.8.8.8 came back with TTL 116, which shows the packet crossed the real internet successfully.

What we confirmed

- Both cores formed a Full OSPF relationship with the FortiGate.
- Both firewalls are Synchronized in the HA monitor.
- Hosts in VLAN 10 and 50 reached 8.8.8.8 with no loss.
- Traffic left through ISP1 (Vodafone), the primary, as expected.

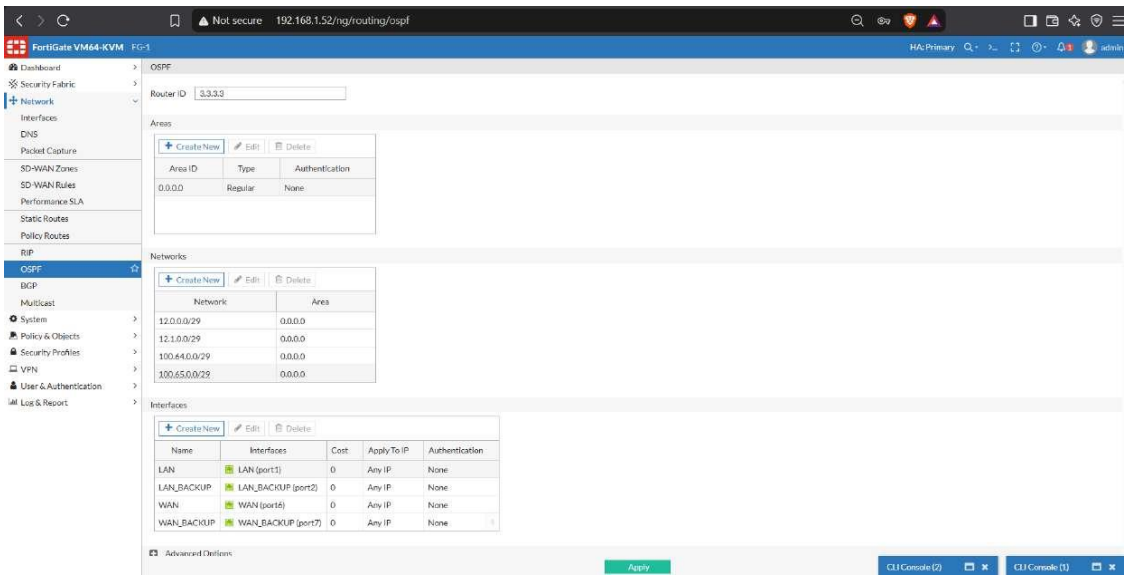
Chapter 12: Problems We Hit and How We Solved Them

These are the most valuable part of the project for review:

Problem	Cause	Solution
HA would break on failover	Cables were crossed between the two firewalls	Same port on both units goes to the same switch
The core could not reach the firewall	Routed ports on different subnets cannot share the cluster IP	Access ports in a transit VLAN with an SVI (VLAN 12 and 13)
SVI stayed down/down	The VLAN did not exist in the VLAN database	Create the VLAN in global config and assign the ports
Hosts could not ping	PortFast network type enabled Bridge Assurance and the port went BKN	PortFast edge on host ports
No default route on the firewall	The health check saw both ISPs as dead and withdrew the route	Fixed the cloud so the ISP routers got DHCP and internet
Wrong WAN design	Mask mismatch (/28 vs /29) and OSPF running on WAN ports	Aligned on /29 and removed WAN from OSPF
SD-WAN would not accept the interface	Static routes still referenced the ports	Deleted them, then added the members and one default route

Example: OSPF configured on the WAN by mistake

At first we added the WAN ports (port6 and port7) to OSPF on the firewall. As a result **ISP1 learned our internal routes** (VLANs 10 to 50), which is wrong because the ISP must not know our internal network.



Wrong: WAN ports (port6 and port7) and networks 100.64 and 100.65 added to OSPF

```
ISP1#sh ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, + - replicated route

Gateway of last resort is not set

      10.0.0.0/24 is subnetted, 5 subnets
O        10.10.10.0 [110/12] via 100.64.0.3, 00:00:07, Ethernet2/0
O        10.20.20.0 [110/12] via 100.64.0.3, 00:00:07, Ethernet2/0
O        10.30.30.0 [110/12] via 100.64.0.3, 00:00:07, Ethernet2/0
O        10.40.40.0 [110/12] via 100.64.0.3, 00:00:07, Ethernet2/0
O        10.50.50.0 [110/12] via 100.64.0.3, 00:00:07, Ethernet2/0
      12.0.0.0/29 is subnetted, 2 subnets
O        12.0.0.0 [110/11] via 100.64.0.3, 00:00:17, Ethernet2/0
O        12.1.0.0 [110/11] via 100.64.0.3, 00:00:17, Ethernet2/0
ISP1#
```

Wrong: ISP1 learned VLANs 10 to 50 through OSPF from the firewall

Fix: we removed the WAN from OSPF. The ISP only needs to see the firewall as a gateway, and the firewall reaches the ISP through the static route and SD-WAN.

Chapter 13: Useful Commands for Review

These commands help you review the network or hunt for a problem:

FortiGate

Command	Use
get system ha status	HA state (who is Primary and Secondary)
get router info ospf neighbor	OSPF neighbors and their state
get router info routing-table all	The full routing table
diagnose sys sdwan health-check	ISP state from the health check
diagnose sys sdwan member	SD-WAN members
diagnose sys session list	Active sessions and the NAT
diagnose debug flow trace start 20	Trace a packet inside the firewall (set the filter first)

Core and access switches

Command	Use
show standby brief	HSRP state (Active and Standby)
show ip ospf neighbor	OSPF neighbors
show ip route ospf	Routes learned by OSPF
show spanning-tree	STP port states (FWD, BLK or BKN)
show vlan brief	VLANs and their ports
show etherchannel summary	Port-channel state
show interfaces trunk	Trunks and allowed VLANs
show port-security	Port security state

ISP routers

Command	Use
show ip interface brief	Interface states and addresses (DHCP on e2/1)
show ip nat translations	The NAT table
ping 8.8.8.8	Test internet reachability

Chapter 14: What Is Left and Project Summary

Not done yet

These tests prove the design is truly redundant. Their results are not recorded yet:

Test	How	Expected
ISP failure	On ISP1 shut down e2/0 during a continuous ping	The ping drops for about 5 to 10 seconds, then returns through WE
Firewall failure	Power off FG-1	FG-2 becomes Primary with the same addresses and the internet continues
Core failure	Power off CORE_1	HSRP moves to CORE_2 and traffic continues

Suggested improvements

- Add port6 and port7 to the HA monitor.
- On ACC-2 change PortFast from network to edge (like ACC-3), and on ACC-1 add switchport mode trunk on the uplinks.
- Write standby 10 ip 10.10.10.254 explicitly on NH_SEC.
- Set the URL filter to Block for every site that should be blocked (Instagram, TikTok and Snapchat were Monitor, Allow and Exempt).
- Add IPS, antivirus and DNS filtering, and enable syslog.

Summary

- The network has four layers, and each layer has its own redundancy.
- The core is split with MSTP and HSRP, so both cores are active.
- The core and the firewall speak OSPF over transit VLANs.
- The firewalls run HA, and internet access is controlled by SD-WAN and a policy with NAT.
- Most of the problems were in the details: cabling, the VLAN database, the PortFast type and the /29 addressing.